

Value-Aware Adaptive Multi-Robot Exploration under Resource Constraints

SOP Progress Review · November 2026

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Since the last review

01

Claim narrowed

Amorín 2019 and WiSER-X added. No more 'first to study stopping'; the claim is about learning how much effort to keep.

02

Problem reframed

From 'when to stop' to 'how much exploration effort is worth buying'. Stopping = zero robots active.

03

Stages A and B done

The problem is real when costs are high; the RL-style target is needed, a big model is not.

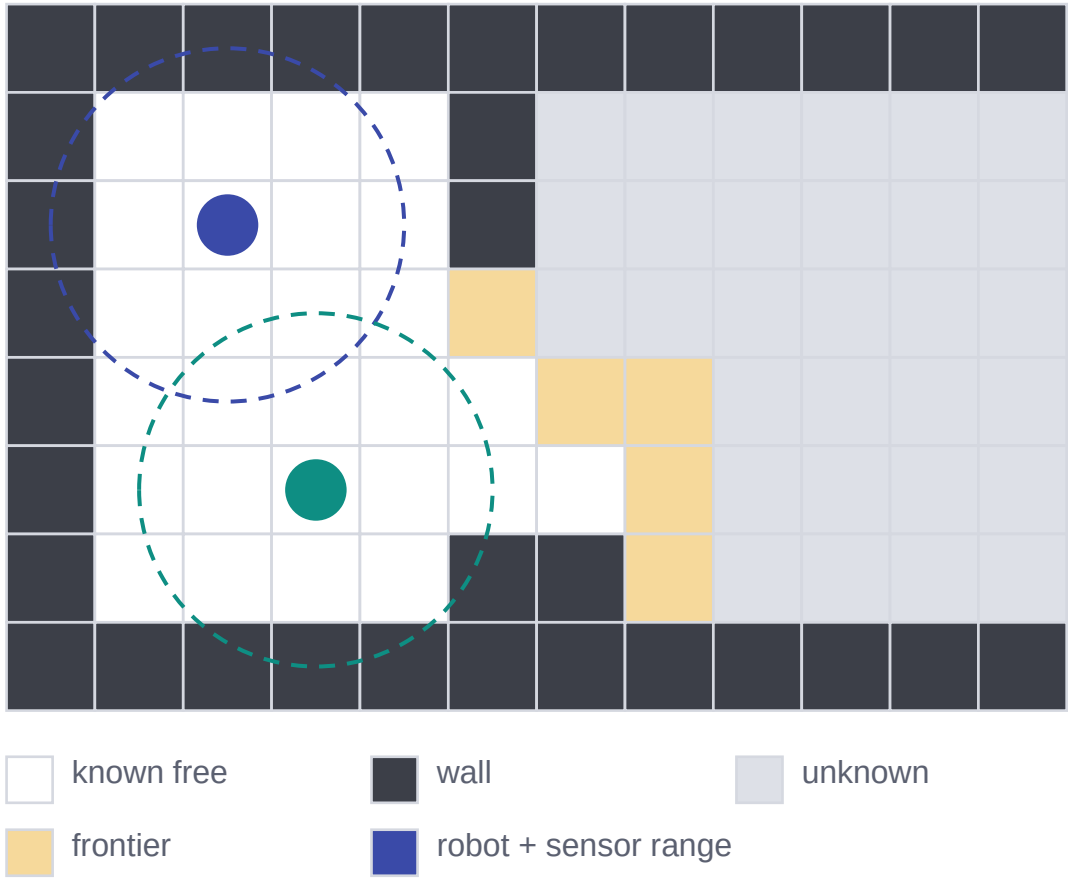
04

Code on GitHub

Simulator, learned stop rule, all experiments and plots, reproducible from saved runs.

Repository: github.com/dev-goyal-chaitanya/marl-simulations (private; access on request)

Multi-robot exploration



Occupancy grid. The map is a grid of cells: free, wall, or unknown. Robots start knowing nothing.

Sensing. Each robot sees cells within a fixed radius, in line of sight. Walls block the view.

Frontier. A known free cell next to an unknown one. Going there reveals new space [1].

Allocation. Each robot gets a different frontier, trading off distance and expected gain [2].

Exploration cycle. Sense → share maps → assign frontiers → move. Repeat.

The open question. After how many cycles should the whole team stop?

How exploration stops today

Stopping rule	How it decides	Used in	What it leaves open
No frontiers left	Explore until nothing reachable is unknown	Yamauchi 1997 [1]; Burgard et al. 2005 [2]	Cost of the last, scattered pockets
Coverage or time budget	Stop at X% mapped or after T minutes	Common in practice and MARL benchmarks	X and T must be re-tuned per map and team
Fleet information gain	Team stops when expected information gain of the fleet is low	Amorín et al. 2019 [3]	Hand-designed criterion; team size not adapted
Uncertainty saturation	Stop when map and pose uncertainty stop improving	Placed & Castellanos 2022 [4]	Single robot; motion and messages not priced
Learned map completeness	A CNN judges if the partial map is complete	Luperto et al. 2026 [5]	Single robot; supervised, not cost-aware
Per-robot return rule	A robot relays data when returning pays more than exploring	Kim et al. 2026 (PRoID) [6]	Per robot, hand-set; mission ends at a deadline
Asynchronous termination	Each robot finishes on its own, without a shared map	Jadhav et al. 2026 (WiSER-X) [7]	Per robot; effort not priced against gain

Common thread: stopping has been studied, but no rule learns, for the whole team, how much exploration effort is still worth its cost.

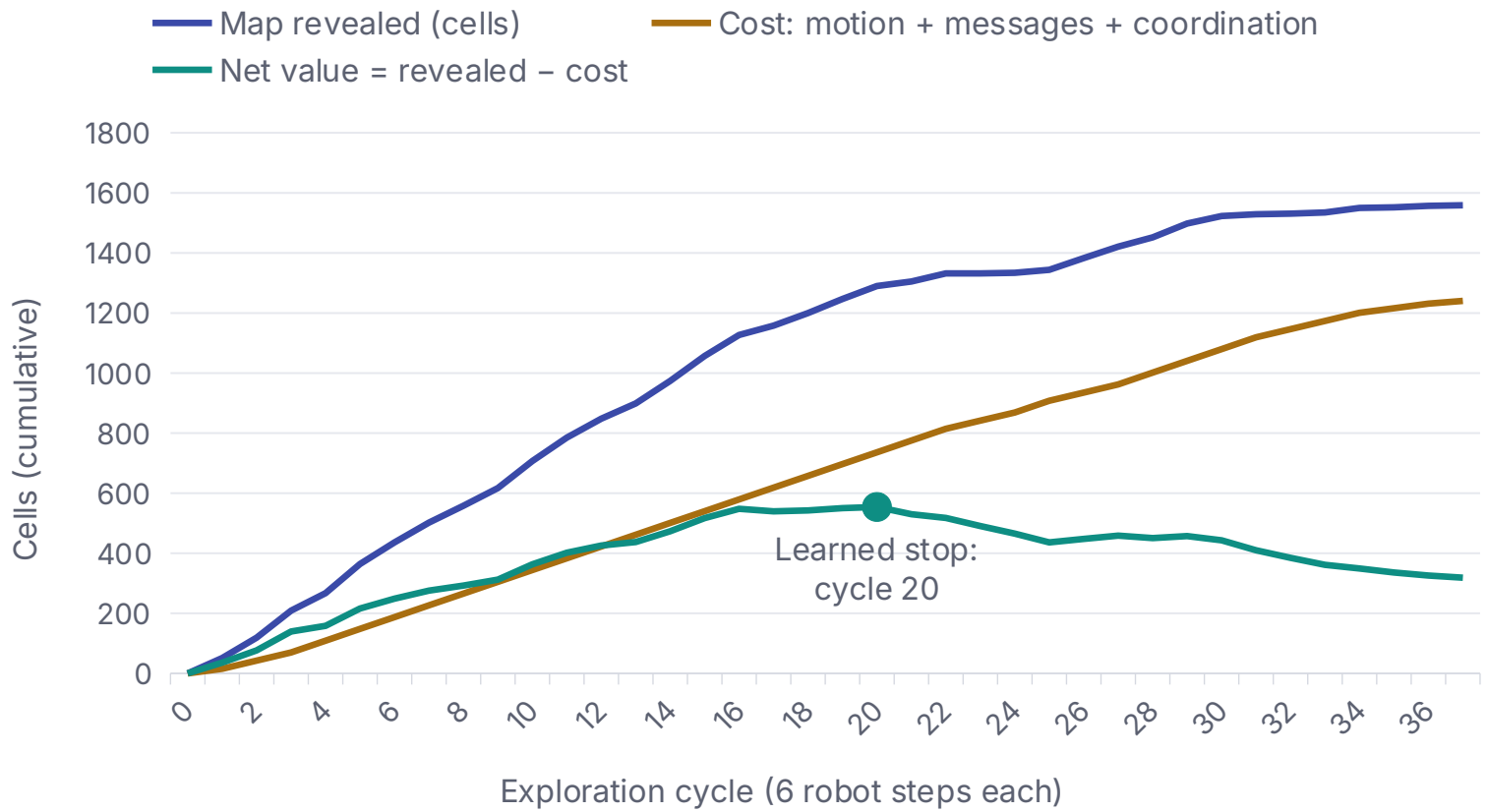
When should the team stop?

Early cycles reveal whole rooms. Late cycles reveal small, scattered pockets that need long drives.

Every cycle keeps costing: robot motion (energy, time), map-sharing messages, re-planning.

So the question is not whether the team *can* explore further, but whether one more cycle is **worth its cost.**

Messages grow as N(N−1) with team size N, so the question gets sharper for bigger teams.



Prototype run, 3 robots, 40×40 map. Stopping at cycle 20 maps 83% of the free space; the last 17 cycles add 17% more map but cut net value from 554 to 319.

Three problems, one contribution

Problem A

Frontier allocation

Which robot explores which frontier?

Well studied: greedy [2], market-based [8], MinPos [9], and recent MARL methods (Tan et al. 2023 [10]; MARVEL 2025 [11]).

Execution substrate

Problem B

Communication

Which robots talk, and what do they share?

Active MARL topic: learned, targeted and bandwidth-limited messaging (TarMAC 2019 [12]).

Modelled as a cost and a constraint

Problem C

Exploration effort

How much exploration (robots, messages) is still worth buying? Stopping = zero effort.

Stop rules exist [1], [4], [5], [6], [3], but none learns how much effort to keep; MARL work fixes the episode length [10], [11].

The contribution of this project

Frontier allocation is the machinery the robots run on, not the claim. **The claim is about how much exploration effort the whole team should keep spending, with stopping as its zero point.**

What is already known

Area	What exists	What is still missing
Multi-robot exploration Yamauchi 1997 [1]; Burgard et al. 2005 [2]; MinPos 2012 [9]; survey [13]	Frontier-based coordination with utility and travel-cost allocation.	Termination is a fixed rule: no frontiers left, coverage or time.
MARL for exploration Tan et al. 2023 [10]; MARVEL 2025 [11]	Learned, decentralised frontier selection; CTDE training.	Episodes end at a fixed length or coverage; stopping is not learned.
Stopping criteria for exploration Amorín et al. 2019 [3]; Placed & Castellanos 2022 [4]; Luperto et al. 2026 [5]; PRoID 2026 [6]; WiSER-X 2026 [7]	Fleet information-gain stop; uncertainty-based and learned single-robot stops; per-robot return and asynchronous termination.	Learning how much effort to keep (active robots, messages) from expected gain vs cost.
Hierarchical MARL Options, Sutton et al. 1999 [14]; Han et al. 2019 [15]	Learned termination of each agent's sub-policy (option).	Termination of the whole task, driven by expected gain versus team cost.
Optimal stopping Tsitsiklis & Van Roy 1999 [16]; Longstaff & Schwartz 2001 [17]; Becker et al. 2019 [18]	Theory and regression or neural stopping rules, mostly single decision maker.	Many agents, partial observability, costs that depend on coordination.
LLM-agent orchestration Hu et al. 2025 [19]; Zhang 2026 [20] (motivation only)	Statistical stop rules for LLM debates; a 2026 preprint survey finds no RL-trained stop.	Same question in a different field. Not part of the proposed system.

Closest prior work, and the gap

Work	Multi-agent	Learned	Decision made for	Adapts team size / messages	Prices motion + messages + coordination
Amorín et al. 2019 [3]	✓	×	whole team (stop)	×	not stated*
Placed & Castellanos 2022 [4]	×	×	single robot	×	×
Luperto et al. 2026 [5]	×	supervised	single robot	×	×
Kim et al. 2026, PRoID [6]	✓	×	each robot	indirectly (robots leave to relay)	time, risk
Jadhav et al. 2026, WiSER-X [7]	✓	not stated*	each robot	×	not stated*
Szostak & Cohen 2024 [21]	✓	✓	partly	×	sampling cost
Han et al. 2019 [15]	✓	✓	sub-policies only	×	×
This project	✓	✓	**whole team**	**planned (Stage D)**	✓

Narrowed claim: to my knowledge, no prior work learns, for the whole team, how much exploration effort to keep buying (active robots, messages), priced against motion, communication and coordination cost. Stopping is the zero-effort case.

** Amorín et al. and WiSER-X characterised from their abstracts; full-text check in progress.*

Research questions

Can a hierarchical multi-agent policy allocate exploration resources (active team size, message budget, and finally termination) according to the expected marginal value of continued exploration?

	Question	Status
RQ1	Can the remaining value of exploration be estimated online from partial observations?	Partly answered: r = 0.88, sign right 89% (Stage B)
RQ2	Can a learned controller reduce exploration effort (robots active) as value declines?	Next: Stage D, this semester
RQ3	Does learning allocation and exploration jointly beat designing them separately?	Next semester: 5-way ablation
RQ4	Does the policy generalise to new environments, team sizes, costs and comms without retraining?	Partly: 36 settings tested; full plan fixed in advance
RQ5	Can robots approximate the central decision with only local observations and messages?	Long term (follow-up work)

Formulation 1: the setting

Symbol	Meaning	In this project
I = {1, …, N}	The agents	N robots (trained with 2, tested with 2 to 8, later 10)
S	True state of the world	True map, robot positions, what has been mapped so far
Ω_i, O	What robot i observes	Cells within 4 cells of the robot, in line of sight
A_i	Robot i's action (Level 2)	Which frontier to drive to, one step at a time
A_o	Team action (Level 1)	Once per cycle: how many robots stay active and the message budget; stop = 0 robots
P	How the state changes	Robots move and sense; the known map grows
R	Reward	Net value of each cycle: cells revealed minus the team's costs (next slide)
b	Belief / team summary	What the team knows at a decision point, compressed into features

Dec-POMDP [22]

The team problem

$$M = \langle I, S, \{A_i\}, P, R, \{\Omega_i\}, O \rangle$$

Dynamics

$$P(s' \mid s, a_1 \ldots a_N)$$

robots move and sense; the known map grows

Observations

$$O(o_1 \ldots o_N \mid s', a_1 \ldots a_N)$$

each robot sees only its own neighbourhood

Our addition: team resource decision (Level 1)

$$a_H(t) = (\ N_active(t),\ B_comm(t)\ )$$

from the team summary b(t); stop ⇔ N_active = 0
(prototype: continue / stop only)

Formulation 2: value of a cycle

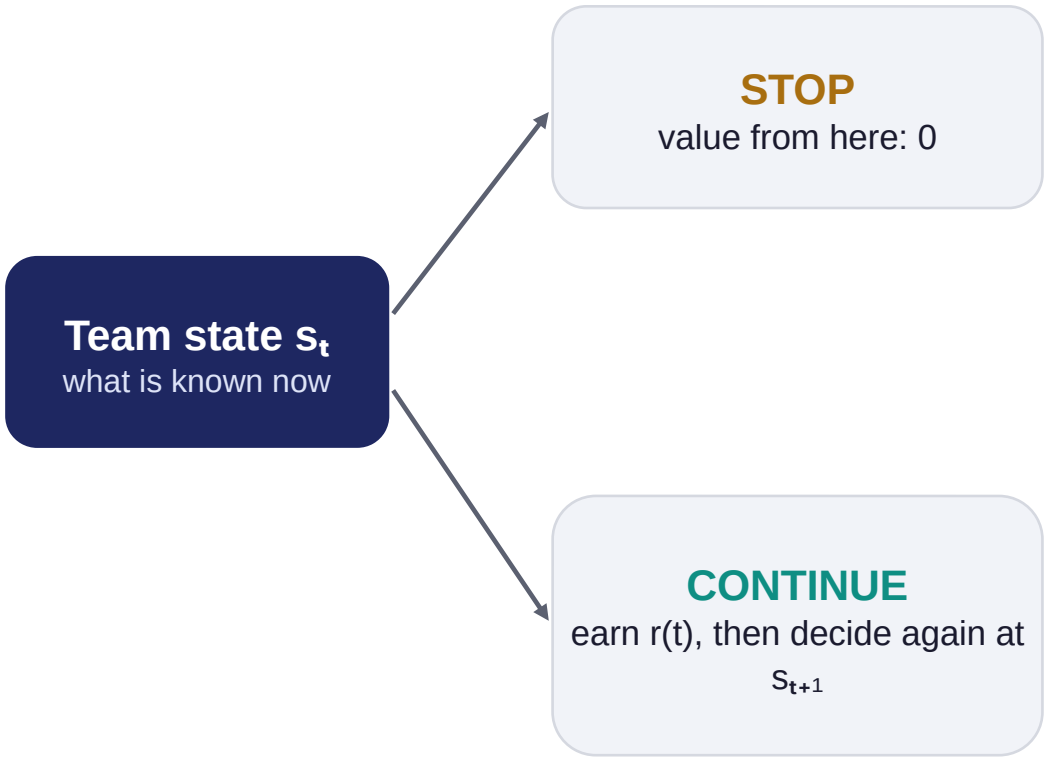
$$r(t) = \Delta I(t) - c_{move} \cdot d(t) - c_{comm} \cdot N(N-1) - c_{coord}$$

ΔI(t)	cells revealed in cycle t (new map)
d(t)	robot steps actually taken in cycle t
N(N−1)	map-sharing messages: every robot to every other
c_coord	fixed re-planning overhead per cycle

Stopping after T cycles earns V(T) = r(0) + … + r(T−1).

Next, from the review: N(N−1) assumes every robot talks to every other. Replace it with the cost over the links actually used, $C_{comm} = \Sigma(\alpha \cdot data + \beta \cdot messages + \eta \cdot latency)$, and measure information as map entropy reduction $H(M_t) - H(M_{t+1})$. Report each part separately, not only the combined score.

Formulation 3: optimal stopping



Value of continuing

$Q(s) = E[\, r(t) + \max(0, Q(s_{t+1})) \mid s_t = s \,]$

max(0, ·): at the next decision the team again picks the better of stop (0) and continue. This is the classical optimal-stopping recursion [16], [17].

Stopping rule

Stop at the first t with $Q(s_t) \leq 0$

i.e. when the expected value of exploring further no longer covers its cost.

Reference for scoring (not a policy)

T* = argmax_T V(T), computed after the run

Needs the future. Every rule is scored as a % of V(T*).

Formulation 4: allocation



Active robots N_active(t) over the mission. Effort shrinks as the remaining value falls; 0 active robots is termination.

What changes

Action: (N_active, B_comm) instead of {continue, stop}

Utility: map entropy reduction, not raw cell counts

Costs: motion, communication, coordination, and later risk

Prototype rule 'stop when Q ≤ 0' = the case with two actions

Marginal value of exploration

$MVE(b_t, a) = E[\Delta U \mid b_t, a] - E[C_motion + C_comm + C_coord + C_risk \mid b_t, a]$

expected extra map utility minus expected cost of choosing effort level a

Decision

$a^* = \text{argmax over a of } Q(b_t, a), \quad Q(b, a) = E[\, r(a) + \gamma \cdot V(b_{t+1}) \,]$

$\gamma \leq 1$ discounts the future; the prototype uses $\gamma = 1$

Termination

$Q(b, STOP) = 0$; stop when max over $a \neq STOP$ of $Q(b, a) \leq 0$

the optimal-stopping idea [16], [17], embedded in resource allocation

How the stop is learned

1 Collect

300 training maps, 2 robots, 32×32. Explore each to the end and log every cycle.

2 Re-score

Score every run under 4 random cost settings (cost randomisation): 1,200 runs.

3 Fit Q

25 rounds of fitted Q-iteration [23]; each round an extra-trees regressor (80 trees).

4 Decide

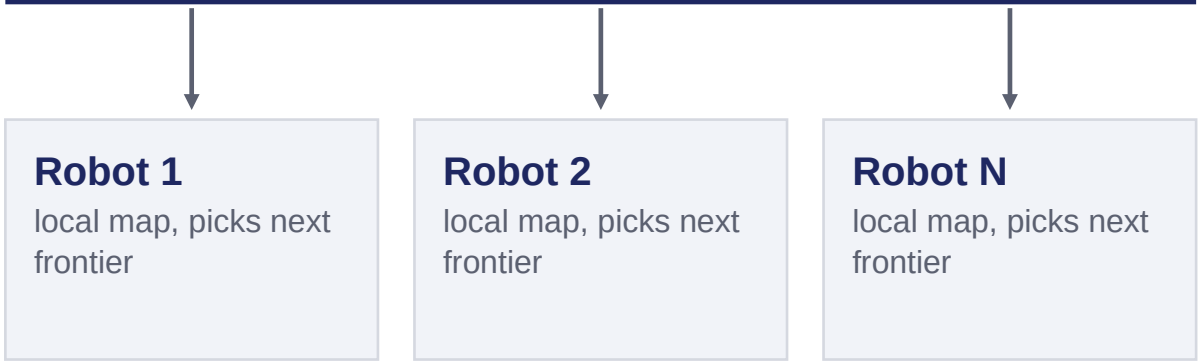
At run time compute Q from the 7 features. Stop when Q ≤ 0.

Feature (all ratios, so they transfer)	Meaning
Map known	Fraction of all grid cells already known
Expected gain ÷ cycle cost	Unknown cells near the planned targets, per unit of cost
Last gain ÷ cost, previous gain ÷ cost	How well the last two cycles paid off
Distance to targets ÷ 6	How far away the planned frontiers are
Share of robots with a target	How much of the team still has work
Frontier cells ÷ map side	How much frontier is left

Two-level architecture

Level 1: mission-level resource controller

Once per cycle. Sees a team summary (map known, expected gain vs cost, recent gain, frontiers). Decides how many robots stay active and the message budget; stopping = 0 robots.



Level 2: robot-level exploration, one shared policy per active robot, every step (planned: MAPPO [24]; QMIX [25] as backup)

CTDE [22], [24], [25]: trained with the full map and all robot states; executed with each robot's own observations. That is what makes it multi-agent hierarchical RL [14], [15].

Why a hierarchy now

A binary stop does not need hierarchical MARL: regression or a threshold could do it.

Choosing how many robots stay active does: the top level sets the effort, slowly; the bottom level must adapt to whichever robots remain.

Today

Level 1 learned (continue / stop); Level 2 a fixed nearest-frontier rule.

Next

Level 1 picks the team size (Stage D); Level 2 learned with MAPPO only if Stage D shows a real gain (Stage E).

Baselines for comparison

Stopping rule	Stops when…	Parameter	Idea
Explore until no frontiers	no reachable frontier is left	none	Classical exploration [1], [2]
Fixed budget	B cycles have run	B = 19 (tuned)	Time budget [10], [11]
Coverage threshold	known fraction of the grid ≥ τ	τ = 0.84 (tuned)	Area threshold [10], [11]
Myopic rule	the last cycle lost value (r < 0)	none	Stop at the first bad cycle; simplified look-ahead rule [26]
Gain-vs-cost rule	α × expected gain ÷ cycle cost < 1	α = 1.6 (tuned)	Hand-made value test, inspired by [2], [6]
Learned value-aware stop	estimated Q ≤ 0	learned	Looks at the whole remaining future [16], [17], [23]
Best in hindsight (oracle)	best cycle, chosen after the run	needs the future	100% reference, not a policy

With α = 1.6 the gain-vs-cost rule stops once expected gain < cycle cost ÷ 1.6. Being added: fleet information-gain threshold [3], map-completeness rule [5], gain-per-time rule [6].

Evaluation metrics

Net value, % of best

V(T) ÷ V(T*), totals over all maps

Main score: did it stop at a good time?

Regret

V(T*) − V(T), as % of the map [27]

Value lost; stable even when V(T) is tiny*

Stop-time error

T − T* in cycles; share early / late

How and when it gets it wrong

Operational cost

Coverage, cycles, robot steps, messages

What the stop saves, and what it leaves

Value estimate quality

Correlation of Q with hindsight value; sign accuracy

Does it know the remaining value?

Robustness

Worst setting, interquartile mean [28], settings ≥ 80%

Does it hold without retraining?

Statistics: paired comparison on identical maps, 95% bootstrap intervals, at least 5 training seeds [28]. Added from the review: Pareto front of map value vs cost as cost weights vary; once robot motion is learned, two oracles (best stop on this run vs best joint policy).

Experimental programme

Stage A

Is the problem real?

Value lost by exploring to the end, across maps, teams, costs

Done

Stage B

Is it learnable?

Simple models vs fitted Q-iteration for the remaining value

Done

Stage C

Stopping benchmark

Robot motion fixed; all stop rules incl. [5], [6], [3] and the oracle

In progress

Stage D

Adaptive team size

8 → 6 → 3 → 1 → 0: same map value at lower cost?

Next

Stage E

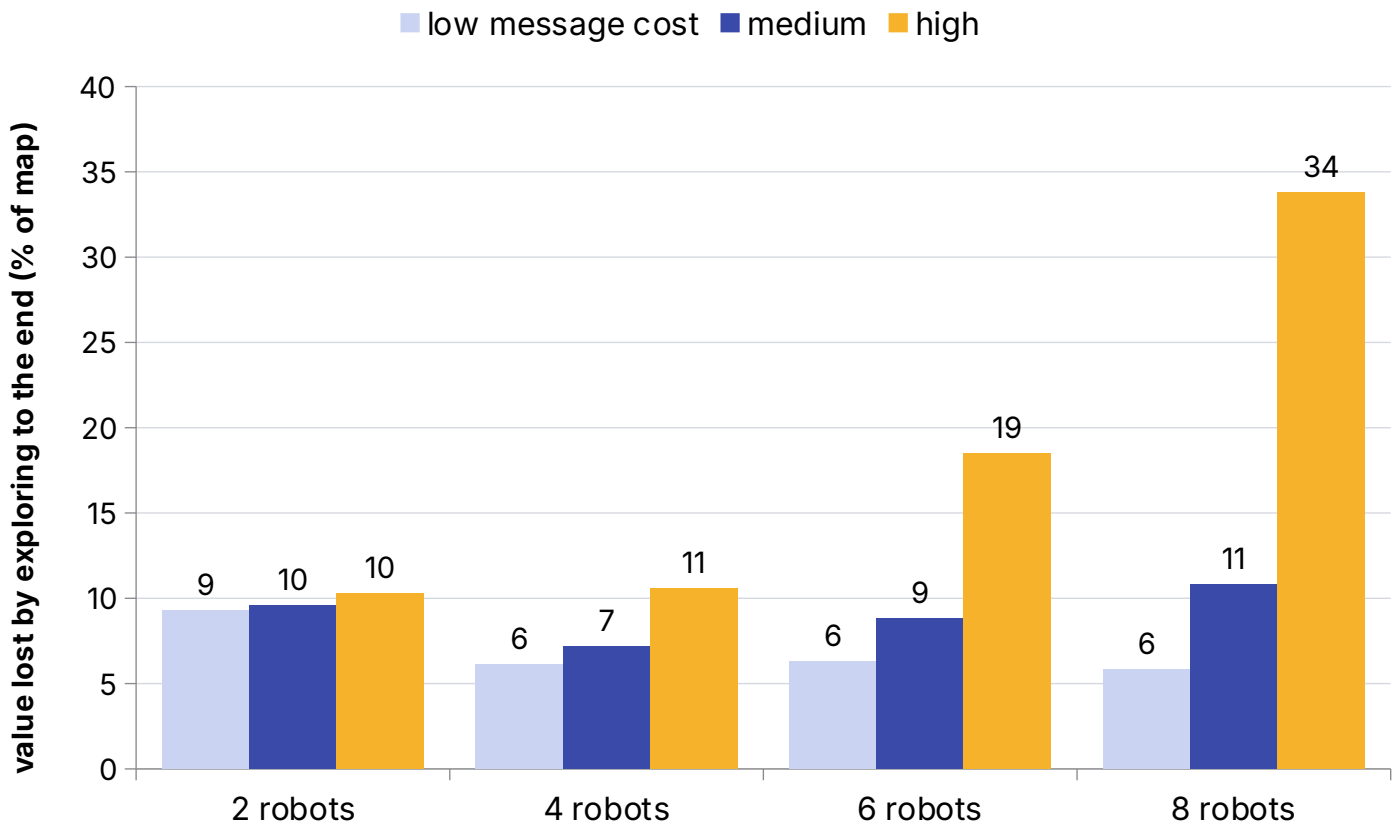
Hierarchical MARL

Resource controller + MAPPO exploration [24]; ablation of 5 combinations

Next semester

Gazebo / ROS2 (MRTA testbed) is used to test what the grid world leaves out: pose noise, sensor noise, message drops, battery variation, robot failure. It is a check, not the contribution.

Stage A: is the problem real?



Averaged over 32/48/64 maps, 40 maps each. On average 11.4% of the map's value and 36% of all driving are spent after the best stopping point.

But it depends on the price of driving

Cost per step	Value lost (% of map)	Driving after best stop
0.5	1.0	4.8%
1	2.2	9.6%
2 (default)	**9.1**	**33%**
3	27.3	68%

Medium message cost, all 12 team/map combinations.

Cheap driving: just explore everything. The work matters when motion or messages are expensive, so the next step is real cost values or a full sweep (Pareto).

Stage B: is RL needed?

Training target	Model	Avg value lost (% map)	Worst setting	Settings < 50%	Sign of value right
fitted Q-iteration	**extra trees (current)**	**3.6**	**9.9**	**1**	**89%**
fitted Q-iteration	random forest	3.7	10.3	1	89%
fitted Q-iteration	linear	3.5	13.0	4	89%
fitted Q-iteration	neural net	4.8	12.1	1	89%
hindsight labels	linear	6.7	30.4	6	85%
hindsight labels	neural net	5.4	37.4	5	79%
hindsight labels	random forest	9.0	35.7	8	80%
hindsight labels	extra trees	9.6	39.0	8	79%
reference: coverage threshold (tuned)	—	4.7	25.3	5	—

Why: hindsight labels assume the team will later stop perfectly, so they overvalue continuing and stop too late (the perfect-foresight bias known in optimal stopping [17]). The model barely matters: even linear works.

Stage C: stopping benchmark

Stop rule	% of best (training setting)	Avg value lost, 36 settings	Worst setting (% of map)
Learned value-aware stop	**91**	**3.6**	**9.9**
Coverage threshold (tuned)	92	4.7	25.3
Gain-vs-cost rule (tuned)	89	6.9	15.8
Fixed budget (tuned)	91	9.1	37.1
Myopic (stop after a losing cycle)	68	11.4	24.5
Explore until no frontiers	74	11.4	44.6

Typical case: a tie with the tuned coverage threshold.

Worst case: 9.9% vs 25.3% lost, with no re-tuning per setting.

Known weakness: 2 robots on 48/64 maps stop 20–33 cycles early; cause not yet diagnosed. One training seed so far.

Stage D: adaptive team size

How it will work

1. Each cycle the controller picks N_active ∈ {0, …, N}.

2. Retired robots stop moving and messaging: no motion or message cost.

3. Training data: runs where robots retire at random times, so every team size is seen.

4. Same fitted Q-iteration, with N_active as an input; pick the size with the highest Q.

Compare

Full team + learned stop · fixed schedule 8 → 4 → 2 → 0 · learned resizing

Success = nearly the same map value at clearly lower motion and message cost.

Questions for you

1. Realistic costs: take energy per metre and bandwidth from the MRTA robots, or report a sweep?

2. A retired robot: park in place, or return to base (and pay that trip)?

3. Retirement permanent, or can robots be re-activated?

4. Paper 1 (value estimation + termination, no MARL) as the first target?

Generalisation plan

Shift	Train → test	Status
Environment	rooms → maze, warehouse, cluttered, subterranean-style	Only rooms so far; adding maps
Scale	32×32 → 64×64 → 128×128	Tested up to 64×64
Team size	train 2–4 robots → test 5, 6, 8, 10	Trained on 2, tested on 2–8
Communication	full → limited radius, bandwidth, packet loss, delays	Only message cost varied
Costs	sweep of motion / message weights (Pareto front)	3 message costs; motion sweep in Stage A

Target result: a policy trained on small teams and maps transfers without retraining to larger teams and unseen environments, keeps a high share of the hindsight-best value, costs clearly less than full exploration, and beats tuned stopping baselines across cost and communication regimes. The numbers come from the experiments, not set in advance.

Plan for this semester

When	Build	Output
Nov, week 1–2	Real cost values or sweep; baselines from [5], [6], [3]	Full Stage C table
Nov, week 2–4	Team resizing (Stage D)	Learned resizing vs fixed schedule vs full team
Late Nov	New map types, 5 seeds, diagnose 2-robot failure	Generalisation table
Early Dec	Write-up of Stages A–D	Report + Paper 1 draft outline

Literature (in parallel)

Shared database: 33 entries today across exploration, stopping criteria, MARL exploration, hierarchical MARL, optimal stopping and communication. Target: 50 by end of November, 80 before the MARL stage.

Each paper recorded on the review's 10 questions: decision made, learned or hand-designed, single or multi-agent, individual or mission-level stop, utility, costs, comm constraints, dynamic team size, centralised or not, test settings.

Possible publication path

Paper 1

Learning the Marginal Value of Continued Multi-Robot Exploration

Continuation-value estimation, value-aware termination, full stopping benchmark, generalisation. No MARL needed. Stages A–C.

This semester

Paper 2

Hierarchical Resource-Aware MARL for Adaptive Multi-Robot Exploration

Which robots explore, how many stay active, when to stop. Stages D–E.

Next semester

Paper 3

Decentralised Value-Aware Mission Termination under Communication Constraints

Local estimates, learned communication, robustness to packet loss and failures. RQ5.

Follow-up

Framing: exploration as the sequential allocation of exploration resources under diminishing returns; termination is the zero-resource boundary.

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Thank you

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